

RESEARCH ARTICLE

Role of microtopography in the expression of soil propagule banks on reclamation sites

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The legacy propagule banks of salvaged topsoils are excellent sources of plant propagules for reclamation of mine sites; however, earlier studies show that less than 50% of the species found in original propagule banks actually establish. We hypothesize that the expression of this legacy propagule bank is limited by a lack of diversity of microsites and appropriate growing conditions. In an operational-scale field experiment we manipulated topographical characteristics and substrate materials and explored early vegetation establishment on an east- and south-facing slope. Three different site treatments with different microtopographic characteristics: (1) leveled surface, (2) parallel ridges, and (3) large loose piles were created using salvaged upland and lowland forest floor soil materials. Placing materials in loose hills and providing heterogeneity in substrates more than doubled plant abundance, species richness, and increased the proportion of species that require higher soil moisture. Lower micro-elevations and northern microaspects of the microtopographic treatments had up to three times higher soil water content and double the species richness. However, the overall slope aspect had a modulating effect on microtopographic positions. The greatest treatment effect of micro-elevation and microaspect on species richness and plant abundance occurred on the east-facing slope, while the greatest effect of treatment on the proportion of species requiring higher soil moisture conditions was observed on the south-facing slope. Variability in microtopography and substrate can create favorable growing conditions at an operational scale and help express a wider range of species from the legacy propagule bank.

Key words: forest restoration, microrelief, seed bank, slope aspect, soil water, vegetation establishment

Implications for Practice

- By creating small hills from loosely piled salvaged forest floor materials, restoration practitioners can substantially increase species richness, plant abundance, and the number of species over a greater range of soil water conditions compared to standard operations, while reducing equipment hours and costs.
- Slope aspect altered the effects of site preparation techniques, and should be accounted for when selecting techniques for achieving specific revegetation objectives. The examined hilled treatment positively affected plant abundance and species richness on the east-facing slope, while increasing the proportion of species requiring higher soil moisture on the south-facing slope.
- Lower microtopographic positions provided moister, more favorable conditions compared to elevated positions, and should be prioritized for planting or seeding to achieve greater revegetation success.

Introduction

Natural forests often provide a broad range of microsites that vary in size, substrates, and climatic and edaphic conditions that result in heterogeneities in the abundance and composition of vegetation. Much of that variation is driven by variables such as topographic heterogeneity at the landscape and microsite

scale, and is modified by factors such as canopy composition and closure, and coarse woody debris (Beatty 1984; Cornett et al. 1997). Plant species vary widely in their prerequisites for establishment, growth, and survival (Harper & White 1974). Greater heterogeneity in resource availability and conditions (i.e. microsite diversity) in a given area increases the likelihood that resource requirements of a greater array of species can be satisfied, and should lead to increased overall species richness and diversity (Nichols et al. 1998; Bartels & Chen 2010).

Surface mining creates severe disturbances in forest regions, as it requires the complete removal of surface soil and overburden material that cover the resource. After mining activities have ceased, many mined areas require the reconstruction of forests including facilitating the restoration of basic ecological functions (Macdonald et al. 2012). While planting trees remains a key aspect contributing to restoring ecological function within

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the post-disturbance forest landscape, the establishment of a diverse understory is considered an essential part of the recovery of forest ecosystem functions (Jacobs et al. 2015; Macdonald et al. 2015). In many forest ecosystems, the understory plant community is one of the greatest contributors to ecosystem species richness, where it can provide a range of functional types and traits that support all trophic levels (Nilsson & Wardle 2005; Gilliam 2007).

Establishing species-rich plant communities in reclaimed forest understories is a challenge that requires the availability of a diverse array of plant propagules (seeds and vegetative fragments) in combinations with the appropriate growing conditions (Macdonald et al. 2012, 2015). Salvaged forest soils placed on reclamation sites can be a valuable source of native plant propagules (Tacey & Glossop 1980; MacKenzie & Quideau 2010), but establishment success of the array of species contained in this material is determined by the specific growing requirements of each species (i.e. the species-specific recruitment niche) (Harper & White 1974; Dalling & Hubbell 2002). Water and nutrient availability, and climatic conditions such as soil and air temperature are important factors influencing seed germination and establishment (Harper & Benton 1966; Silver-ton 1999). Of those factors, adequate water availability is a crucial driver of germination and early establishment success for many species (Rodriguez-Iturbe 2000; Bhattacharjee et al. 2008). Particularly at higher latitudes, water availability is also affected by topographic position and aspect that greatly influence energy inputs due to low sun angles (Holland & Steyn 1975), especially when a moderating tree canopy is absent (Chen et al. 1993). Microrelief or surface topography are also important factors for vegetation establishment (Klug-Pümpel 1982; Frouz et al. 2011) and the presence of sheltered microsites benefits germination and establishment of plants from seeds by protecting them from high respiration rates, early and late frosts, and herbivory (Elmarsdottir et al. 2003; Schott et al. 2014).

Compared to natural forests, forest reclamation sites provide a unique opportunity to explore some of the fundamental hypotheses that relate plant richness to microsite heterogeneity and resource availability, because the climatic and edaphic impacts on the initial resource availability are mostly independent of historical site conditions in these constructed landforms. In addition, reconstruction of these sites allows for the creation of a variety of landscape and microscale topographical features through variations in soil surface topography (microtopography) (Biederman & Whisenant 2011) and substrate types (Hutchings et al. 2003; Ojekanmi & Chang 2014). The role of soil surface topography in the re-development of forest understory communities has not yet been explored in boreal forest reclamation settings, but this approach has been successfully applied in temperate forest, wetland, and grassland restoration projects (Ewing 2002; Bruland & Richardson 2005; Biederman & Whisenant 2011; Frouz et al. 2011). It is possible that increasing microtopographic variability on upland forest reclamation sites may prove to be a valuable strategy for facilitating ecosystem recovery and species diversity.

In this study, we explored the impact of microsite heterogeneity on the expression of the propagule bank that is contained in

salvaged and placed surface cover soils in reclamation areas. We hypothesized that on upland forest reclamation sites, surface soil treatments that provide a greater range of microtopographic and substrate variability will conserve higher species richness, plant abundance, and a greater range of species functional types relative to that originally contained in the propagule bank. The specific objectives of this research were (1) to evaluate how slope aspect and microtopographic surface treatments will affect the early expression of the initial propagule bank (i.e. abundance, richness, and early community compositions) and explore these responses in relation to edaphic conditions (i.e. snow accumulation, soil water availability, bulk density, and soil temperature) at the site scale and (2) to examine in more detail the effect of microtopographic position (micro-elevation and microaspect) of microtopographic features on the propagule bank expression, and explore this response to soil water availability, bulk density, and soil temperature at the microsite level.

Methods

Study Area

Research was conducted on an oil sands mining lease (Shell Albion Sands) located 70 km north of Fort McMurray, Alberta, Canada (lat. 57°15'N, long. 111°23'W) within the Central Mixedwood natural forest subregion (Natural Regions Committee 2006). The average annual temperature is 0.2°C, with monthly January and July temperatures of −17.4 and 17.1°C, respectively. Average annual precipitation and potential evapotranspiration is 465 and 518 mm, respectively (Marshall et al. 1999). Snow fall (November–April) is about 30% of annual precipitation, and the average frost-free period is 97 days. During the year of the study, 2015, the average annual daily temperature was 3.4°C and average daily temperature over the growing season (15 May to 31 August) was 17.5°C. The total annual and growing season precipitation in 2015 was 26% (310 mm) and 55% (126 mm) lower than the long-term average, respectively (Environment Canada 2015, Fig. S1).

Study sites were established on a south-facing and east-facing slope of a large hill landscape feature that was constructed from the overburden material overlaying the ore. The landscape feature was contoured in the summer of 2014, and subsequently capped with two different salvaged cover soils: a peat mineral mix (PMM) and forest floor material (FFM). PMM is a mixture of peat and some of the underlying mineral soil salvaged from black spruce (*Picea mariana*) dominated lowland forests, with the mineral content varying greatly among PMM materials depending on salvage depth. The FFM is a mixture of upland organic forest LFH (litter-fibre-humic) soil horizons and underlying mineral A and B soil horizons salvaged to a depth of about 30 cm. The upland forest stands were dominated by jack pine (*Pinus banksiana*) with lesser components of trembling aspen (*Populus tremuloides*) and white spruce (*Picea glauca*). The understory was dominated by dwarf shrubs (*Vaccinium* spp.) and lichen (*Cladonia* spp. and *Cladina* spp.). The soil was an Eluviated Dystric Brunisol that developed in this area on course-textured, acidic,

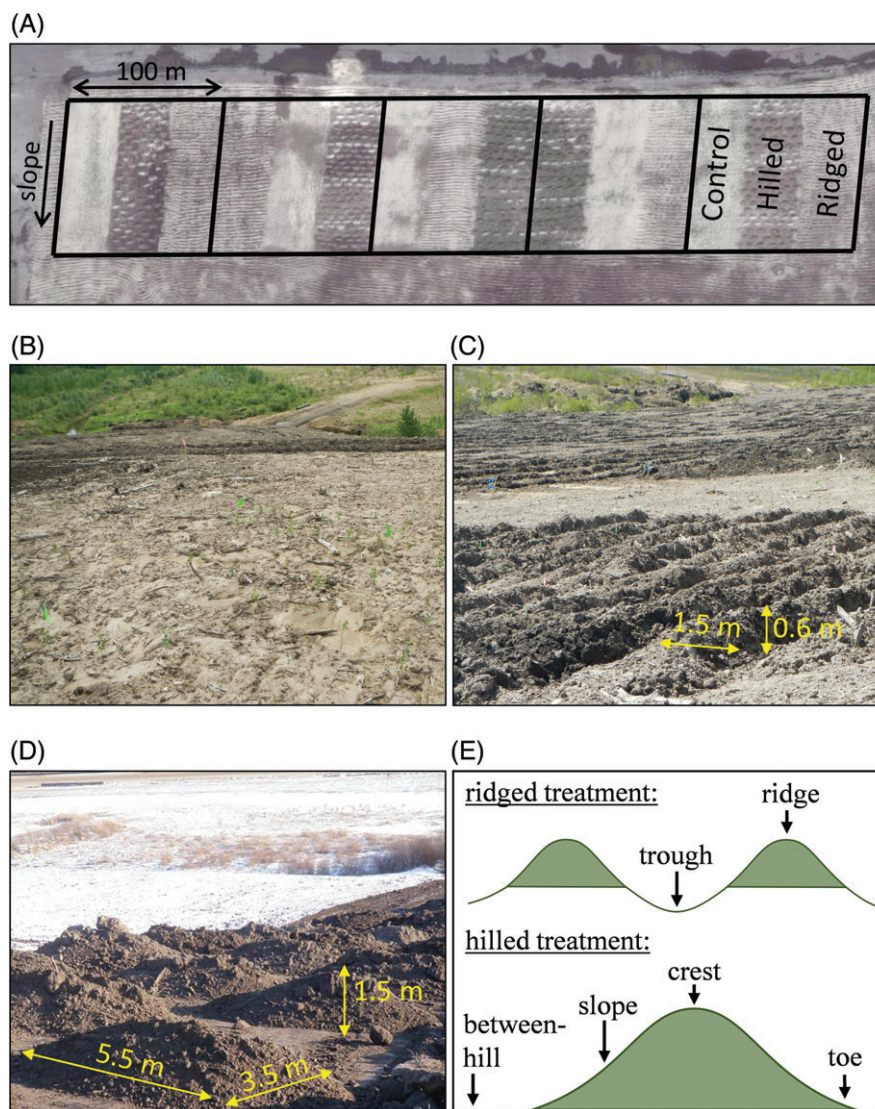


Figure 1. Aerial view of one of the two slope aspects containing five blocks, with the control, hilled, and ridged microtopographic treatments replicated in each block (A); photographs of the control (B), ridged (C), and hilled (D) treatment plots; and diagrammatic representation of the micro-elevation positions identified on micro topographic features in the ridged and the hilled treatments (E).

glaciofluvial outwash parent materials. The organic forest floor was approximately 5 cm thick and of a mor humus form. Both the PMM and the FFM were salvaged in April 2014 and temporarily stockpiled in several large piles until late fall of 2014.

Study Design

After placing and leveling a 35-cm base layer of PMM as a cover soil over the constructed landscape feature in October of 2014, five blocks (100 × 100 m each) were established on the south-facing and the east-facing slopes of the landscape feature (Fig. 1). In each block, three microtopographic treatments (control, ridged, and hilled) were randomly assigned to 30 × 100 m areas with a 3-m buffer between them (Fig. 1A). In the control treatment, a 15-cm layer of FFM was placed over a 35-cm PMM base layer and leveled using a D6 Caterpillar®

bulldozer, which is a representation of the current operational practice (Fig. 1B). The ridged treatment was constructed in the same manner as the control, but tracks of a D8 Caterpillar® bulldozer were used to create deep surface soil ridges by driving the machine perpendicular to the slope in parallel lines (Fig. 1C). To minimize the compaction of the base layer, this was done in early spring (March 2015) when the underlying overburden and the PMM layer were still frozen while the surface FFM layer had already thawed. Ridges were 0.4–0.8 m high, 1.5 m wide, and located approximately 1–2 m apart. The hilled treatment was constructed after the base layer had frozen. Using a D6 Caterpillar® bulldozer, stockpiled PMM or FFM cover soils were selected randomly and pushed from the upper slope to the lower slope and placed as unconsolidated hills into off-set rows evenly distributed in each treatment plot (Fig. 1D). The hills were made of either FFM or PMM (randomly placed across

the plot), had approximately oval bases (approximately 3.5 m wide and 5.5 m long), were 1.5 m tall, and spaced approximately 1.5 m apart.

Vegetation and Edaphic Variable Assessment

For the microtopographic treatment-level vegetation assessment, 15 vegetation sampling quadrats (1 × 1 m) were randomly placed along an 80-m transect running parallel with the slope and located in the center of each microtopographic treatment plot (total of 30 transects). The location of each quadrat relative to the position (micro-elevation) on a microtopographic feature in the hilled and the ridged treatments was also noted (i.e. between-hill, toe, slope, and crest in the hilled treatment, and trough and ridge in the ridged treatment) (Fig. 1E). At the microtopographic treatment level, all possible micro-elevation positions were represented in each transect of each plot to allow the overall effect of microtopographic position within the microtopographic features to be evaluated.

A more detailed analysis was also undertaken to explore the impact of microaspect and cover soil material on early seedling establishment at a finer scale. This was done only in the hilled treatment plots. Three FFM hills and three PMM hills were selected in each plot and eight sampling quadrats (0.5 × 0.5 m) were established on each hill. On each hill, quadrats were arranged in the four cardinal directions, with four quadrats each located at the lower slope position (i.e. hill toe) and upper hill slope position.

Vegetation was assessed in early- to mid-August 2015. In all sampling quadrats (i.e. 1 m² and 0.25 m²) plants were identified to species and the number of individuals of each species was determined. Soil water content and soil temperature were measured in the center of each vegetation sampling quadrat three times during the growing season (May 16, June 9, and August 10, 2015) using a hand-held mobile TDR (time domain reflectometry) probe (POGO, Stevens Water Monitoring Systems, Inc., Portland, OR, U.S.A.). To assure the accuracy of the water content values, the TDR probe was calibrated for each cover soil (Appendix S1).

Snow surveys were conducted during the first growing season to determine how microtopographic treatments and the overall slope aspect influenced snow accumulation, redistribution, and the resulting snow water equivalent (SWE). The surveys took place between 17 February and 20 February, 2015 when the accumulated snow pack was near its peak (Fig. S2). Snow depth data were collected in three replicates of only the control and the hilled treatment on the south- and east-facing slope, as the ridged treatment had not yet been constructed. Snow depth was measured with a ruler every 2 m on either side of a 100-m-long transect placed in the center of each treatment plot (30 transects) parallel with the direction of the slope. Snow density (g/cm³) was determined at three randomly located points for a range of snow depths (at 1-cm depth intervals) on each slope aspect. Snow density was not correlated to snow depth on either slope aspect (both $p > 0.79$). SWE (g/cm²) was calculated by multiplying average snow density by each snow depth on the respective slope aspects (Elder et al. 1998; Jonas et al. 2009).

Soil bulk density was measured at three random locations per microsite position per treatment plot: three random samples from the control treatment plots, three samples each from the ridge and trough positions in the ridged treatment, and three samples each from the slope position (all south aspect) of FFM hills and PMM hills, and from the between-hill position in the hilled treatment plots. Soil bulk density samples were collected by hammering a cylindrical metal ring of known volume into the soil (inside diameter 7.22 cm, height 7.75 cm, volume 317.3 cm³), carefully extracting it, then drying and weighing the sample (Blake 1965).

Initial Propagule Bank of Salvaged FFM and PMM Cover Soils

To explore the inherent propagule bank (seed and vegetative bud bank) of the two different cover soils used in this study, 18 bulk soil samples were collected in August 2014 from the salvaged PMM and FFM stockpiles. Each bulk soil sample consisted of three soil cores (3.3 L, 14.5 cm diameter, 20 cm tall), with two cores collected from the top 20 cm of the stockpile surface and a third taken at 20-cm depth.

Plant propagules in the soil samples were concentrated to approximately 30% (3 L) of the original bulk sample to increase the likelihood of germination/sprouting as outlined by Ter Heerdt et al. (1996). Concentrated samples were spread onto a weed free potting soil mix (Sunshine LA4 mix potting soil with 55–65% sphagnum peat, perlite, dolomitic limestone, gypsum, wetting agent), and placed in plant trays (51 × 26 cm) at a 2-cm thickness. Rhizomes and other vegetative propagules separated during sieving were placed and covered in a layer of potting soil mix and incubated separately. The trays were incubated in the greenhouse at a 16-hour photoperiod with day and night temperatures set to 21 and 15°C, respectively, and relative humidity set between 50 and 75%. The trays were watered every 1–3 days using an automatic misting system to maintain moist soil conditions. As soon as plants could be identified, they were counted and removed from the incubation trays; plants that initially could not be identified were transplanted into separate pots to allow for further development. After new plants stopped appearing in the seed trays, irrigation was paused for 1 week to allow the soil to dry slightly and was disturbed by lightly raking the soil surface to encourage maximum possible germination (Ter Heerdt et al. 1996). The incubation of the propagule bank study was terminated after 16 weeks when no new germinants were observed. Since the propagule banks yielded very low numbers of viable vegetative fragments, the number of germinants and sprouted vegetative propagules were combined for each bulk sample for the subsequent analysis.

Data Analysis

All data analyses were performed in The R Project for Statistical Computing (R Core Team 2015). Prior to performing analysis of variance (ANOVA) procedures, datasets were tested to ensure that assumptions of normality and equality of variances of the residuals were met. Post-hoc pairwise comparisons of least-squares means with no p -value adjustment were then

performed for each significant main effect and interactions to determine which treatment combinations were different from the others.

Propagule Bank Study. The propagule bank study in the greenhouse was set up as a completely randomized design with cover soil as the treatment effect (FFM and PMM). The experimental unit for the propagule bank study was a tray (containing the bulked subsamples). The emerged plant species were assigned into one of three functional types associated with disturbance (includes ruderal and grassland species), forests, and high-moisture habitats. The assignment was based on the preferred habitat types of each species described in the Flora of Alberta (Moss & Packer 1983, Table S1). Non-metric multidimensional scaling analyses using Bray–Curtis dissimilarity coefficients based on the number of individuals of each species was performed to visualize the differences between the communities in the two cover soils and the three microtopographic treatments. A one-way MANOVA was also performed to compare the communities in the two cover soils and the three microtopographic treatments. Species area saturation curves constructed using the specaccum function (Oksanen et al. 2016) and fitted by reciprocal function (Colwell & Coddington 1994) indicated that potentially 78 and 82% of the species were detected in the FFM and PMM cover soil, respectively (Fig. S3) and that this study sufficiently represents the actual number of species in the propagule bank.

Slope Aspect and Microtopographic Treatments. To determine the effect of microtopographic treatments and slope aspects on response variables, two-way ANOVA was used on transect species richness, plant abundance, proportion of functional type, soil water content, SWE, and soil bulk density at the treatment scale with the data being averaged at the treatment plot-level (transect) (Table S3). Plant abundance was transformed with $\log_{10}$ and species richness was transformed with Box-Cox transformation using packages car and geOR (Fox & Sanford 2011; Ribeiro & Diggle 2016) to achieve normality. For soil temperature, the effect of microtopographic treatments separately within a given slope aspect (One-way ANOVA) could be examined, as operational constraints allowed only one aspect to be measured in a given day. Soil temperature was averaged at the treatment plot level and blocked in time to minimize bias due to changing temperatures over the day (Table S3). Indicator species characterizing each of the three field treatments were determined based on their indicator values (Dufrêne & Legendre 1997) at $p < 0.01$ by using labdsv package (Roberts 2016).

Microelevation and Microaspect Study. Two-way ANOVAs were performed to explore the effect of micro-elevation position with slope aspect on quadrat species richness, plant abundance, soil water content, and soil bulk density (Table S4). The influence of microelevation position was analyzed as a randomized block design, and thus, hilled and the ridged microtopographic treatments were analyzed separately. Each of the 15 experimental quadrats along the transects in the ridged treatment was classified by its microtopographic position: trough or ridge. In

the hilled treatment, the 15 quadrats along each transect were classified by their microelevation positions: between-hill, toe, slope, and crest. These positions were used for all measured variables apart from soil bulk density measurements, which were only collected in the between-hill and slope positions on either a FFM or PMM hill. All the measured variables were averaged for each microelevation position in each treatment plot and the data were analyzed as a split-block design. In the ridged treatment, slope aspect (south-facing and east-facing) and micro-elevation (trough and ridge) were the two fixed factors. In the hilled treatment, slope aspect (south-facing and east-facing), cover soil (FFM and PMM) and microelevation (between-hill and slope for soil bulk density; between-hill, toe, slope, and crest for the other variables) were the three fixed factors. In the hilled treatment, plant abundance was transformed with square-root transformation.

In the hilled microtopographic treatment, a more detailed study of microaspect was analyzed as a randomized split-block design with slope aspect (south-facing and east-facing), cover soil (FFM and PMM), and microaspect (N, E, S, W) as three fixed factors. For this analysis, the quadrats located on the upper and lower slope of each hill were combined for each cardinal direction. Three-way ANOVAs were performed for quadrat species richness, plant abundance, and soil water content (Table S5). The residuals of all three variables met the assumptions of normality and equality of variances, and no transformations were required.

Results

Species and Community Composition

The two cover soils explored in the greenhouse study differed in their number of species and abundance of propagules in the initial propagule bank. The propagule bank of the FFM cover soil material contained a total of 213 individuals (205 individuals from seed and 8 individuals from vegetative fragments) while the PMM cover soil material contained a total of 1,439 (1,433 individuals from seed and 6 individuals from vegetative fragments). The average propagule density in the FFM was 1.2 ± 1.2 (SD) propagules/L, while the propagule density in PMM was on average 8.0 ± 8.2 (SD) propagules/L. Overall, the FFM and PMM cover soils contained 21 and 28 species, respectively. The FFM and PMM cover soils had similar numbers of species that were forest-associated (8 for FFM and 11 for PMM; see species assignment to functional types in Table S1) and disturbance-associated (10 for FFM and 8 for PMM). However, the PMM cover soil contained more species associated with high soil moisture (three for FFM and nine for PMM). When combining the initial FFM and PMM propagule bank, the total initial seedbank had 35 species, of which 7 were unique to the FFM cover soil and 14 were unique to the PMM cover soil. This is in contrast to the total of 58 species that were found in the field study after the first growing season, of those 23 species (40%) had also been present in the initial propagule bank.

The hilled treatment had the highest species richness compared to the ridged and the control (Table 1). While the three

Table 1. The total number of species in the three microtopographic treatments, and of those the number of species that were either contained in the initial propagule bank, were unique to, or shared among treatments.

	Number of Species						
Microtopographic Treatment	total	contained in propagule bank	unique to treatment	shared among treatments	associated with high moisture	associated with forests	associated with disturbance
Control	39	12	4	23	4	15	19
Hilled	47	12	11	24	9	20	18
Ridged	37	10	2	25	5	15	17

microtopographic treatments had similar numbers of species that either originated from the initial propagule bank found in the greenhouse study (approximately 11 species) or were shared among the three treatments (approximately 24 species), the hilled treatment had more species unique only to this treatment (11 species) compared to two and four for the other two treatments (Table 1, species unique to treatment listed in Table S2). The hilled treatment contained the highest number of species associated with high-moisture habitats and forests, while the number of species associated with disturbance was similar among the three microtopographic treatments (Table 1).

Based on species richness and abundance, the early community composition differed among the three microtopographic treatments and the initial propagule banks of the two cover soils ($p < 0.001$) (Fig. 2). The plant communities developed from the initial PMM and FFM propagule banks were different from each other and also from the communities observed in three microtopographic treatments in the field ($p < 0.001$ for all). However, within microtopographic treatments, the early plant community in the hilled treatment differed from the communities in the ridged and the control ($p = 0.001$ for both), while the latter two had similar early community composition ($p = 0.123$). The indicator species analysis revealed that the field treatments were dominated by disturbance-associated native and introduced species such as *Chenopodium berlandieri* and *Melilotus alba*, while the initial propagule bank of FFM

cover soil was dominated by forest-associated species such as *Rubus idaeus*, and the initial propagule bank of PMM cover soil was dominated by moisture-loving species such as *Ranunculus sceleratus*.

Slope Aspect and Microtopographic Treatments

Average species richness and plant abundance varied with microtopographic treatment, but not with slope aspect; however, there was a slope aspect by treatment interaction (both $p < 0.030$) (Table S3). The hilled treatment had higher species richness on the east-facing slope compared to the south-facing slope, while there was no difference between slope aspects in the control and ridged treatments (Fig. 3A & 3B). On the east-facing slope, average species richness and plant abundance was higher in the hilled treatment compared to the control and the ridged treatments. A different trend was observed on the south-facing slope, where species richness was highest in the control and lower in the ridged treatments, with no difference in plant abundance between microtopographic treatments.

The proportion of species associated with high soil moisture contained in total species pool was also influenced by the microtopographic treatments and not slope aspect, but with a significant slope aspect by treatment interaction ($p = 0.013$) (Fig. 3C; see also Table S3). The control treatment had a lower proportion of species that require higher soil moisture (moisture-loving) on the south-facing slope compared to the east-facing slope, while there was no impact of slope aspect in the ridged and hilled treatments (Fig. 3C). However, on the east-facing slope, the proportion of moisture-loving species was highest in the hilled compared to the ridged treatment, while on the south-facing slope the control had the lowest proportion of moisture-loving species (Fig. 3C). The proportion of forest- and disturbance-associated species was unaffected by the slope aspect and the microtopographic treatments (all $p > 0.08$, see Table S3).

Average soil water content differed with microtopographic treatment ($p = 0.001$), but not slope aspect ($p = 0.889$). There was also a slope aspect by treatment interaction ($p = 0.047$). On the east-facing slope, average soil water content was higher in the hilled treatment compared to the control and the ridged treatment, while on the south-facing slope soil water content did not differ among microtopographic treatments (Fig. 3D). The slope aspect and microtopographic treatments had no impact on average snow accumulation (both $p > 0.138$); however, it needs to be considered that snow accumulation was only measured in the

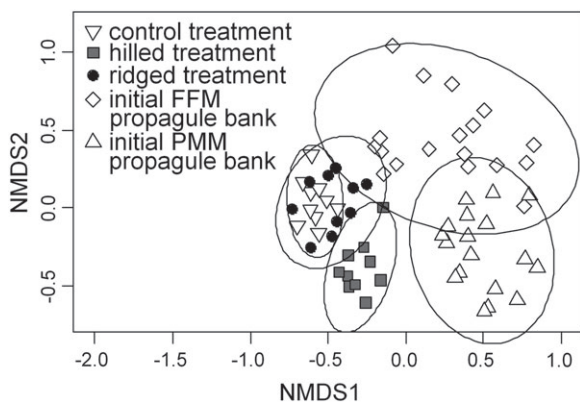


Figure 2. NMDS analysis comparing the community composition of the initial FFM and PMM propagule bank in the greenhouse with the three microtopographic treatments after one growing season in the field. The same FFM and PMM cover soil materials were used in the greenhouse and in the field study. Ellipses indicate 95% probability of distributions.

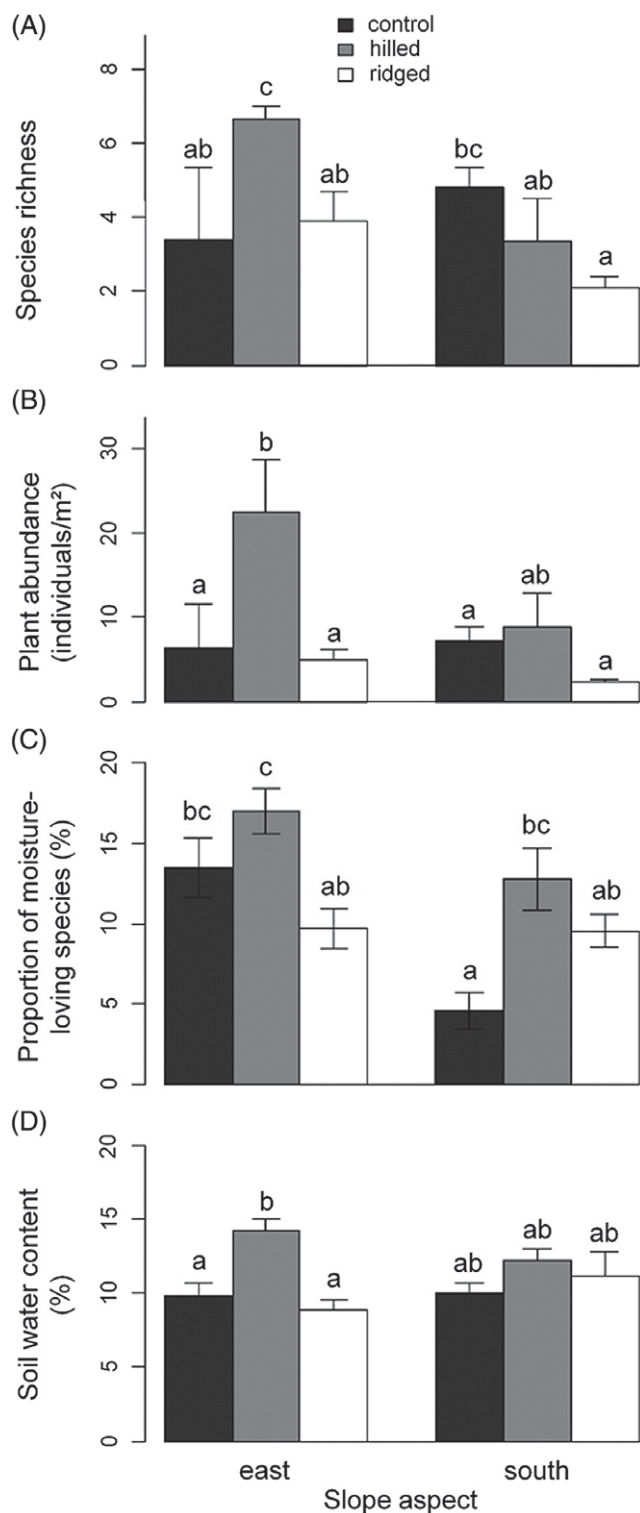


Figure 3. Average species richness (A), plant abundance (average number of individuals per m²) (B), proportion of species that are associated with high soil moisture (moisture-loving) over the total species number (C), and soil water content (D) in the three microtopographic treatments within the two slope aspects ($n = 5$ transects). Bars with different letters indicate differences among treatment means. Error bars indicate $\pm$ SE (A, B, D) and $\pm$ SE (C) ($\alpha < 0.05$).

hilled and control treatments (see Methods). Soil temperature was not affected by microtopographic treatments ($p = 0.880$) while average soil bulk density differed among the microtopographic treatments ($p < 0.001$). The control treatment had the highest average bulk density, followed by the ridged treatment and the hilled treatment with 1.45 ± 0.02 , 1.34 ± 0.02 , and 1.03 ± 0.02 g/cm³, respectively.

Microelevation and Microaspect

In the ridged treatment, microelevation position (ridge vs. trough) affected average species richness, plant abundance, soil water content and soil bulk density (all $p < 0.03$; see Table S4A). In the ridged treatment, species richness and plant abundance was higher in troughs (3.40 ± 0.64 spp./m² and 4.30 ± 1.04 individuals/m², respectively) than on ridges (2.33 ± 0.41 spp./m² and 2.49 ± 0.53 individuals/m², respectively). Soil water content and soil bulk density was also higher in troughs (12.70% and 1.38 ± 0.03 g/cm³, respectively) than on ridges (5.09% and 1.30 ± 0.02 g/cm³, respectively). Soil temperature was not affected by the microelevation position within the ridged treatment (data not shown).

In the hilled treatment, the different microtopographic positions (Fig. 1E) had an effect on species richness, plant abundance and soil water content (all $p < 0.017$; see Table S4B). Here, species richness and plant abundance was higher at the toe and the between-hill position compared to the slope and crest position (Fig. 4A & 4B). Soil water content was also highest in the between-hill and the toe position ($p < 0.001$, Fig. 4C). Soil temperature was not affected by hill microelevation position ($p = 0.999$).

In the hilled treatment, the three-way ANOVA indicates an effect of both microaspect and cover soil on species richness, plant abundance, and soil water content (all $p < 0.04$) (Fig. 5). A slope aspect by cover soil by microaspect interaction was observed for species richness ($p = 0.037$, Fig. 5A). Overall, species richness was greatest on hills constructed with PMM compared with hills made from FFM on the east-facing slope, while on the south slope the two cover soils had no impact on species richness. On the east-facing slope, the north and south microaspects had higher species richness when hills were constructed with PMM cover soil material, while the west microaspect had the lowest richness when hills were constructed with FFM cover soil. However, species richness was not different among microaspects on the south-facing slope regardless of cover soil material. For plant abundance, no interaction was observed between microaspect and slope or cover soil, thus microaspect was plotted separately (Fig. 5B). Greater plant abundance was observed on north microaspects than on the west microaspects, while plant abundance on south and east microaspects was not different from both the north and west microaspects ($p = 0.004$, Fig. 5B). Plant abundance also varied with cover soil ($p < 0.001$), and there was slope aspect by cover soil interaction ($p = 0.003$). On the east-facing slope, plant abundance was higher on PMM hills than on FFM hills, while on the south-facing slope plant abundance was similar between the two cover soils (data not shown).

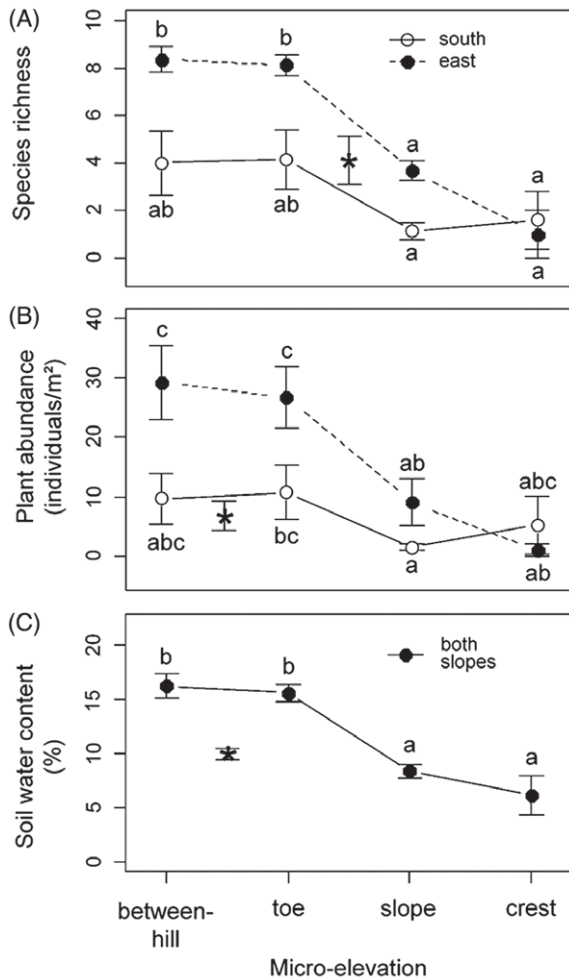


Figure 4. The average species richness per m² (A), and number of individuals (B) in response to four microelevation positions (between-hill, toe, slope, crest) by slope aspect (east and south slope aspects shown separately due to significant microelevation position-slope aspect interactions) within the hilled treatment ($n = 5$). The soil water content (C) in response to four microelevation positions, averaged across both slope aspects (east and south slope aspects are combined, as there was no significant microtopographic position-slope aspect interaction) ($n = 10$). For comparison, a star symbol was added to indicate the average for these response variables in the control treatment. Bars with different letters indicate differences among treatment means. Error bars indicate $\pm$ SE ($\alpha < 0.05$).

Soil water content differed with microaspect, with a slope aspect by microaspect interaction ($p < 0.001$; Fig. 5C). On the east-facing slope, soil water content was higher on the west compared to the east and south microaspects, while the north microaspect of hills located on the south-facing slope had higher soil water content compared to the east, south and west microaspects. Soil water content also varied with cover soil, with a slope aspect by cover soil interaction ($p = 0.04$) (not shown). On the east-facing slope, soil water content was higher on PMM hills compared to FFM hills, while on the south-facing slope, soil water content did not differ between the two cover soils.

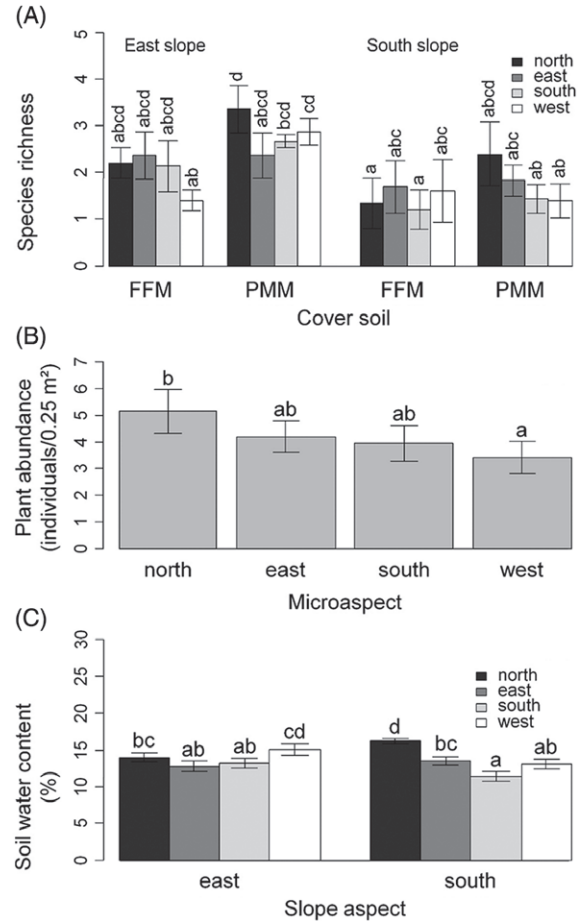


Figure 5. Average species richness (A) in response to slope aspect, cover soil, and microaspect ($n = 5$); average plant abundance (B) in response to microaspect ($n = 10$); and average soil water content (C) in response to slope aspect and microaspect ($n = 5$). Bars with different letters indicate differences among treatment means. Error bars indicate $\pm$ SE ($\alpha < 0.05$).

Discussion

The results of this study indicate that the creation of diverse microsite conditions in reclamation areas is a valuable tool in forest restoration that can improve the diversity of species initially establishing on the site. We had hypothesized that the increased microtopographic heterogeneity and the range of microclimatic conditions and gradients result in increased early species richness and establishment. Since the hilled and the ridged treatments provided greater microtopographic heterogeneity we expected increased richness in both; however, only the hilled treatment with the selectively placed hills appeared to create a range of conditions that was conducive to greater species richness and plant abundance early on. Beneficial for forest redevelopment and restoration, this greater species richness also resulted in plant communities with overall larger numbers of early and later successional forest associated species that were different from the ridged and control treatments (Table 1). This treatment also provided favorable (i.e. relatively higher soil moisture) conditions on the harsher and drier south-facing slope, which allowed for the establishment

of proportionally more species that prefer higher soil moisture conditions compared to the other treatments. This is in contrast to the somewhat reduced topographical variation provided by the ridged treatment, which had little or no positive effect on plant abundance or species diversity and was often comparable to the control. Our results agree with those from other studies (Dhillion 1999; Biederman & Whisenant 2011; Frouz et al. 2011), which showed that naturally occurring and constructed mounds had a positive effect on the species richness and abundance of plants in the restoration of temperate forest and grassland ecosystems, and also increased the growth and survival of artificially seeded species (Ewing 2002). Further, increasing microtopographic variability has also been shown to increase vegetation richness in wetlands by providing unique conditions that suit a wider range of species (Bruland & Richardson 2005; Lieffers et al. 2017).

Although both the hilled and the ridged treatments increased microtopographic heterogeneity, several important differences appeared to have encouraged more effective overall vegetation expression and recruitment of a larger range of plants in the hilled treatment. The effectiveness of the microtopographic features might have been influenced by their size and shape, the soil placement technique, and the location of the contrasting cover soil materials. The larger size of the microtopographic features in the hilled treatment likely resulted in greater range of growing conditions with more sheltered microsites favorable for early plant establishment. The size of the microtopographic features might have also influenced the capture of long-distance dispersed seeds from sources adjacent to the site (Titus & del Moral 1998; Farrell et al. 2012), as a large percentage of the species found on the sites were not detected in the initial seed bank. Casual observations in the field revealed that the loose placement of the cover soil in the hilled treatment could have resulted in greater fine-scale microtopographic heterogeneity, which have been shown to increase species diversity in restored grasslands (Deák et al. 2015). Additionally, the loose placement of soil, the presence of exposed PMM at the surface, and limited heavy-machinery passes in the hilled treatment, compared to the control and the ridged, resulted in the lower bulk densities measured in this treatment. This is consistent with previous research, which has shown soil compaction from one or two additional bulldozer passes can significantly reduce vegetation expression and regeneration (Zenner et al. 2007; Mundell et al. 2008). However, while the ridged treatment also had overall lower average soil bulk density than the control, plant abundance and richness was similar or even lower than the control. The poor performance of the ridged treatment may be due to high soil bulk density found in the troughs, which were comparable to the control. This and the potential of shedding of water from the ridges into the troughs might have led to high water accumulation in the troughs during spring snowmelt and sporadically through the growing season. Casual observations in the field revealed that there was very little or no standing water in the control and the hilled treatments, while the troughs contained standing water in early spring and after heavy rainfalls, which could have adverse effects on germination and early establishment of germinants arising from the propagule bank.

Aside from the lower soil bulk density, the loose and rough placement of cover soil in the hilled treatment resulted in greater exposure of PMM base layer compared to the other two treatments, and increased germination success of the propagules that occur in PMM cover soil. The hilled and the ridged treatments were expected to have higher soil water content because of increased snow accumulation and redistribution as a result of the microtopographic features (Elder et al. 1991). However, in our study the microtopographic features did not result in higher average snow retention, and therefore did not contribute to higher soil water content via snow accumulation. The higher soil water content observed in the hilled treatment was likely the result of higher infiltration capacity of the loosely placed soil resulting in reduced run-off (Appels et al. 2011), and a greater exposure of PMM with its higher water holding capacity (Walczak & Rovdan 2002) at the soil surface. Lastly, the increased overall soil water availability in the PMM and between-hill positions were likely a major factor in facilitating the establishment of species that require higher moisture availability. At an early stage, forest reclamation sites are treeless and the surface soil is exposed to higher evaporation rates than forested stands with a canopy (Chen et al. 1993; Geiger et al. 2009; von Arx et al. 2013). This exposure was further aggravated in this study due to the below-average precipitation during the sampling year. Adequate soil moisture levels that are crucial during seed germination and initial seedling development (Harper & White 1974; von Arx et al. 2013) might not have been reached, which is supported by the limited expression (65%) of the initial propagule bank in the field study.

Microelevation Positions and Microaspect

Elevational differences in the hilled and ridged treatments, and microaspects in the hilled treatment created heterogeneity that was reflected in the increased spatial variability in species and plant abundance within the treatments. Interestingly the effect of microelevation in the hilled treatment was more pronounced on the east-facing slope, while the effect of microaspect in the hilled treatment was more pronounced on the south-facing slope. The spatial heterogeneity in species richness and plant abundance was likely driven by differences in edaphic conditions modulated by climate. Although vegetation expression is strongly tied to soil water content (Harper & White 1974; von Arx et al. 2013), establishing plants often benefit from sheltered conditions occurring with microelevation or microaspect even if soil water content remains unchanged (Elmarsdottir et al. 2003; Schott et al. 2014; Valenzuela et al. 2016). For example, the low-lying between-hill and toe microelevation positions in the hilled and the trough position in the ridged treatments had not only higher soil water content but were likely more sheltered from the sun and wind compared to microelevation positions located on the slope and crests of the hills and ridges. On the other hand, the impact of microaspect on richness and abundance in the hilled treatment was greatest on the north microaspects particularly on the south-facing slopes; the west microaspects did poorly most likely due to the greater sun exposure at the warmest part of the day. Further, soil water

content was the highest on the north microaspect of hills on the south-facing slope, while the west microaspect was the wettest on the east-facing slope. It is not clear if these differences in soil moisture are due to accumulated runoff from up slope and/or shading. Nonetheless, it appears that plant abundance not only increased in response to high soil water content (highest on upslope-facing microaspects: that is, west on the east-facing slope and north on the south-facing slope) but also the reduced solar radiation input (lowest on the north-slope aspect) on these exposed sites.

Overall the results of our study indicate that the effect of topographical variation on plant abundance and the richness of species will also depend on the overall slope aspect, which would have implications for landscape feature construction and the use of microtopographic treatments in forest restoration. Even though the differences in edaphic conditions that we measured on both slope aspects were relatively small (likely a result of the limited monitoring or an overall dry year), the level of environmental stress was most likely higher on the south aspect where the energy input is greater particularly at higher latitudes (Holland & Steyn 1975). This suggests that increased microtopography should be more valuable on sites that are exposed to greater stress conditions (i.e. south-facing vs. east-facing slopes in northern latitudes). However, in our study the positive effect of the hilled treatment on the overall plant abundance and species richness was highest on the east-facing slope. This might indicate that the greater stress conditions experienced on the south-facing slope reduced the overall impact of the topographical treatment on plant abundance and species richness. This observation is supported by earlier work, which suggests that the effect of microrelief on vegetation can be negligible at low-productivity sites with high environmental stresses, is positive at moderate-stress sites, and could potentially have negative effects at low-stress sites (Loneragan & del Moral 1984). The results of this study highlight the importance of creating and providing sheltered and mesic microsites during the reconstruction and restoration of forests, particularly when establishing forest-associated understory species from an existing soil propagule bank.

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Supporting Information

The following information may be found in the online version of this article:

Appendix S1. Calibration of TDR probe.

Table S1. Species assignment to functional types, and their presence and dominance.

Table S2. A list of species absent from the propagule bank and unique to each microtopographic treatments.

Table S3. Results of ANOVAs at microtopographic treatment plot level.

Table S4. Results of ANOVAs at micro-elevation level.

Table S5. Results of ANOVAs at microaspect level in the hilled microtopographic treatment.

Figure S1. Daily average air temperature and daily precipitation during the 2015 growing season.

Figure S2. Snow accumulation during 2014/2015 winter period.

Figure S3. Species area curves for the propagule bank expression of FFM and PMM cover soils in the greenhouse.